HOSPITAL OS

Enterprise-Grade Multi-Hospital Management Platform

Complete AI Agent Upgrade Blueprint

Version	Modules
2.0.0 Enterprise	**12 Core Modules**

1. Executive Summary

Hospital OS v2.0 is a full transformation from a single-hospital management tool into a cloud-native, multi-tenant enterprise SaaS platform designed to power entire hospital networks. This blueprint covers every architectural decision, database change, AI integration, and feature build your AI agent needs to execute the upgrade — with zero ambiguity.

The platform will serve three distinct consumer tiers simultaneously:

- Super Admin (Anthropic / Platform Owner): Manages all hospital tenants, plans, billing, and global analytics.
- Hospital Admin (Each Hospital): Manages their own staff, departments, patients, and configurations.
- Patients: Access a self-service portal for AI health assessments, appointments, lab results, prescriptions, and payments.

Core Philosophy
Every feature must be AI-augmented, not just AI-decorated. AI should be deeply woven into clinical workflows.
Patient experience is paramount. From WhatsApp notification to discharge, the journey should be frictionless.
The system must be HIPAA-aligned, audit-complete, and production-hardened from day one.
Multi-tenancy must be enforced at every layer — DB, API, UI, and file storage.

2. Multi-Tenant Architecture Redesign

This is the most critical architectural change. The entire system must shift from a single-hospital model to a multi-tenant model where every piece of data is scoped to an organization.

2.1 Tenancy Model

Use a shared-database, schema-per-tenant approach with a top-level organizationId foreign key on every table. This keeps infrastructure costs low while maintaining strong data isolation. Each hospital becomes an Organization record.

New Top-Level Entities

Organization	Represents a hospital or hospital network. Root of all data. Has plan, config, branding.
SuperAdmin	Platform-level user. Can create/suspend organizations, view aggregate analytics, manage billing.
OrganizationUser	Links a User to an Organization with a role. Users can belong to multiple organizations.
OrganizationConfig	Per-org settings: enabled modules, feature flags, AI model preferences, SMTP, WhatsApp credentials.
OrganizationBranding	Logo, colors, patient portal subdomain, email from-name. Used to white-label patient portal.

Database Strategy for Multi-Tenancy

- Add organizationId: String as the FIRST field on every major model in Prisma schema.
- Create a global Prisma middleware that automatically injects organizationId into every query — no query should ever run without a tenant scope.
- Add composite indexes on (organizationId, id) for all primary lookup tables for performance at scale.
- Use Prisma's $extends feature to create an organization-scoped client factory that wraps all queries.
- Row-level security (RLS) policies should be added when migrating to PostgreSQL for production.

2.2 New Prisma Schema Additions

New Models to Add to schema.prisma
Organization — id, name, slug, plan (FREE|STARTER|PRO|ENTERPRISE), status, config, createdAt

OrganizationConfig — orgId, whatsappApiKey, whatsappPhoneId, smtpHost, smtpUser, smtpPass, openAiKey, razorpayKey, razorpaySecret, enabledModules (Json), aiModel
OrganizationBranding — orgId, logoUrl, primaryColor, accentColor, patientPortalSubdomain, emailFromName, emailFromAddress
SystemAuditLog — extend existing with organizationId, actorRole, actorId, module, action, recordId, diff (Json), ip, userAgent
Notification — id, orgId, patientId, channel (WHATSAPP|EMAIL|SMS), type, payload (Json), status, sentAt, deliveredAt
AiHealthAssessment — id, orgId, patientId, sessionToken, responses (Json), aiSummary, riskLevel, recommendedTests (Json), recommendedSpecialties (Json), createdAt
PatientPortalSession — id, orgId, patientId, token, expiresAt, deviceInfo
AppointmentSlot — id, orgId, doctorId, date, startTime, endTime, isBooked, bookedBy
Payment — id, orgId, invoiceId, patientId, amount, currency, provider (RAZORPAY|CASH|INSURANCE), status, gatewayRef, createdAt

3. Super Admin Panel

The Super Admin panel is a god-mode dashboard that allows the platform owner to manage the entire hospital network. It must be completely isolated from hospital-level panels, running on a separate route (/superadmin) protected by a separate auth layer.

3.1 Authentication

- Separate JWT secret for super admins — never share with org-level tokens.
- Enforce 2FA (TOTP via authenticator app) for all super admin accounts.
- IP allowlisting as optional extra security layer stored in environment variables.
- Super admin sessions expire in 2 hours with sliding refresh.

3.2 Organization Management

Feature: Hospital Onboarding Wizard

When a new hospital is added, a guided multi-step wizard should collect:

1. Hospital name, legal name, registration number, address, contact info.
2. Plan selection: Free, Starter, Pro, Enterprise with feature diff clearly shown.
3. Admin account creation: first admin user email, name, role. System sends invite email with temp password.
4. Module activation: Which modules to enable for this org (Lab, Pharmacy, IPD, Finance etc).
5. Branding setup: Upload logo, set primary/accent colors, choose patient portal subdomain.
6. Integration keys: WhatsApp Business API credentials, SMTP settings, optional OpenAI key override.
7. Confirmation + auto-seed: DB is seeded with default departments, default roles, and a welcome notification.

Feature: Organization Dashboard

- Table of all hospitals with status indicators (Active, Trial, Suspended, Churned).
- Per-org metrics: total patients registered, active admissions, revenue this month.
- Quick actions: Suspend org, Change plan, Impersonate org admin (with full audit trail), View logs.
- Bulk actions: Export all orgs to CSV, send platform-wide announcement emails.

Feature: Platform Analytics

- Aggregate dashboards: total patients across all hospitals, total revenue processed, total lab orders, total prescriptions dispensed.
- Growth charts: new hospitals per month, patient registrations per month.
- AI usage metrics: Total AI assessments run, AI triage calls, cost per org.

- Alert center: Orgs approaching storage limits, failed payment webhooks, AI API errors.

Feature: Plan & Billing Management

- Define feature flags per plan in a JSON config stored in DB.
- Track plan usage: how many users, how many patients, how much storage per org.
- Automated plan enforcement: if org exceeds limits, show upgrade prompt in their admin panel.
- Invoice generation for enterprise clients with PDF download.

4. Hospital Admin Panel

Every hospital gets their own scoped admin panel after login. All data shown is automatically filtered by their organizationId. The hospital admin is the operational owner of their instance.

4.1 Dashboard & Overview

- Live stats: Patients in OPD queue, Active IPD admissions, Pending lab orders, Pending pharmacy orders.
- AI insights widget: AI-generated daily summary of key operational metrics and anomalies.
- Revenue snapshot: Today's collections, pending dues, insurance claims.
- Staff online now: Shows which staff are actively logged in.
- Quick links to critical actions: Register walk-in, Assign bed, View ER queue.

4.2 Staff & Doctor Management

Add & Manage Staff

The admin panel must support full lifecycle management of all staff types:

Role Types	Doctor, Nurse, Receptionist, Lab Technician, Pharmacist, IPD Manager, Finance Officer, Radiologist
Add Staff	Form: name, email, phone, role, department/specialty, shift, join date. System sends invite email with credentials.
Doctor Profile	Enhanced: MBBS/MD/specialization, registration number, consultation fee, languages spoken, available days/slots, photo, bio for patient-facing portal.
Credentials	Admin can reset passwords, revoke sessions, toggle 2FA requirement per staff member.
Shifts	Define shift templates (Morning 8AM-4PM, Evening 4PM-12AM, Night 12AM-8AM). Assign staff to shifts with calendar view.
Leave Management	Doctors can mark leave. System auto-redistributes OPD queue to available doctors in same specialty.
Performance Metrics	Patients seen per day, avg consultation time, patient satisfaction score from feedback.

4.3 Department & Ward Configuration

- Create/edit departments: name, head doctor, room numbers, floor.
- Ward management: Create wards (General, ICU, Maternity, Pediatric, Surgical), assign beds, set per-day room rates.
- Configure OT (Operation Theatre) schedules linked to surgeon availability.

- Bed matrix live view: color-coded grid — Available (green), Occupied (red), Under Maintenance (gray), Reserved (amber).

4.4 System Configuration

- Notification templates: Customize WhatsApp and email templates for appointment reminders, lab results, discharge summaries.
- AI configuration: Choose which AI model powers triage (GPT-4o, Claude, local fallback). Set confidence thresholds for auto-routing.
- Fee schedules: Set consultation fees per doctor/department, lab test prices, room rates. These feed into auto-billing.
- Insurance panel: Add empanelled insurance companies, configure pre-auth workflows, set coverage rules.
- Integrations panel: Connect/disconnect WhatsApp Business API, Razorpay, SMTP server.

5. Patient Portal — Complete Redesign

The Patient Portal is the most patient-facing surface of the entire system. It must be world-class: fast, beautiful, mobile-first, and deeply personalized using AI. Patients access it via a unique subdomain per hospital (e.g., cityhospital.hospos.in).

5.1 Patient Acquisition via AI Health Campaign

Campaign Trigger Flow

This is the innovation that differentiates Hospital OS from every other HMS. Here is the complete flow:

8. Hospital admin creates a campaign from the admin panel: selects target segment (all patients, patients not visited in 6 months, patients with diabetes tags, etc.), selects notification channel (WhatsApp, Email, or both), writes a custom intro message.
9. System sends a personalized WhatsApp or Email message to each patient in the segment. The message includes the patient's name, hospital branding, and a unique one-time-link to start their AI Health Assessment.
10. Patient clicks the link, lands on the AI Health Assessment portal — no login required at this stage, the link contains a signed JWT that identifies the patient.
11. Patient completes a dynamic conversational assessment (15-20 questions). The AI adapts follow-up questions based on prior answers.
12. AI generates a comprehensive health summary report, displayed to the patient with risk indicators, recommended lab tests, and suggested specialist consultations.
13. Patient is prompted to create/log into their portal account to save the report, book appointments, and get follow-up care.

AI Health Assessment Engine — Technical Spec

Assessment Engine Design
Frontend: Multi-step conversational UI built in React. Each question fades in progressively. Progress bar at top.
Backend: Assessment session stored in AiHealthAssessment table with sessionToken. Questions stored as JSON array.
Question Types: Single select, Multi select, Slider (pain scale 1-10), Free text, Date picker (last visit/symptom onset).
AI Prompt Strategy: Compile all answers into a structured JSON payload. Send to OpenAI GPT-4o or Claude API with a system prompt that instructs it to act as a senior physician reviewer generating a pre-consultation summary.
Output Schema: { summary: string, riskLevel: LOW|MODERATE|HIGH|CRITICAL, symptoms: string[], possibleConditions: string[], recommendedTests: [{ testName, reason, urgency }], recommendedSpecialties: [{ specialty, reason, urgency }], lifestyleNotes: string[] }

Fallback: If AI API fails, use a deterministic rule-engine based on symptom-to-specialty mapping tables stored in DB.

Privacy: All assessment data is encrypted at rest. Patient can delete their assessment data at any time.

5.2 Patient Authentication & Onboarding

- New patients: Phone number + OTP (via WhatsApp or SMS) as primary auth. No password friction.
- Existing patients: Phone OTP, or email magic link. No password to remember.
- Post-auth onboarding: Collect full name, DOB, gender, blood group, emergency contact, insurance details.
- Profile photo upload (optional). Stored in cloud storage (S3-compatible) keyed by org + patient.
- Patient gets a unique UHID (Universal Hospital ID) per organization they register with.

5.3 Patient Portal Features

Health Dashboard (Home)

- Greeting with name, upcoming appointments in the next 7 days.
- Saved AI Assessment report with key risk indicators.
- Recent lab results with trend graphs (e.g., HbA1c over 3 months).
- Active prescriptions and refill reminders.
- Outstanding balance and quick Pay Now button.

Book Appointment

- Search by specialty or doctor name. Filters: gender preference, language, in-person vs video.
- Doctor card shows: photo, name, qualification, ratings, consultation fee, next available slot.
- Calendar slot picker: available slots shown in green, unavailable in gray. Time zone aware.
- Booking confirmation: immediate WhatsApp + email confirmation with appointment details and directions.
- Reminder notifications: 24 hours before and 1 hour before the appointment.
- Rescheduling and cancellation self-service with configurable cancellation policy (e.g., free if 24hr notice).

Video Consultation (Telemedicine)

- Integration with a WebRTC solution (Daily.co or Jitsi) for video calls directly in portal.
- Doctor sees patient's AI assessment summary before the call starts.
- During call, doctor can share screen, write prescriptions, and order labs.

- Post-consultation: prescription and lab orders auto-uploaded to patient's portal.

Lab Results

- All lab results listed in reverse chronological order.
- Each result shows: test name, value, reference range, flag (High/Low/Normal), date.
- Trend charts for key biomarkers (CBC, LFT, KFT, thyroid, lipid panel).
- AI interpretation note generated by AI explaining results in plain language (not clinical jargon).
- Download result as PDF. Share with external doctor via secure time-limited link.

Prescriptions & Medications

- List of all active and past prescriptions.
- Each prescription: drug name, dosage, frequency, duration, prescribing doctor, date.
- Drug interaction checker: warns if two current prescriptions have known interactions (powered by OpenFDA API).
- Refill request: Patient can request refill from pharmacy directly from portal. Pharmacist gets notified.

Medical Records & Documents

- Upload and store external records: old reports, scan images, discharge summaries from other hospitals.
- All records organized by date and type. Full-text search.
- Doctor can see uploaded external records during consultation.

Payments

- Outstanding bills listed with breakdown: consultation fee, lab charges, pharmacy, room charges.
- Pay online via Razorpay (UPI, cards, netbanking, wallets).
- View payment history with downloadable receipts.
- Insurance claim status tracking if the visit is under insurance.
- EMI option for large bills above a configurable threshold.

6. Doctor Panel — AI-Augmented Clinical Workspace

The Doctor Panel is where clinical work happens. It must be a zero-friction, information-dense workspace that keeps a doctor focused on the patient, not the software. AI must reduce documentation burden by 60%.

6.1 Doctor Home — Live Queue

- Real-time queue of all patients waiting for this doctor, sorted by triage priority (Emergency first).
- Each queue card: patient name, age, chief complaint, wait time, triage level color badge, AI assessment flag if they have a pre-visit assessment.
- One-click to start consultation. Previous visits dropdown without navigating away.
- AI queue optimizer: suggests reordering based on patient acuity and doctor availability to minimize total wait time.

6.2 Consultation View — EHR

Pre-loaded AI Context

When a doctor opens a patient's consultation, the AI should have already:

- Summarized the patient's last 3 visits into a concise paragraph.
- Flagged any pending lab results since the last visit.
- Highlighted any drug interactions in current medications.
- Pulled the AI pre-assessment summary if the patient completed one.
- Identified any upcoming chronic disease milestones (e.g., HbA1c due in 2 weeks for diabetic patient).

SOAP Notes with AI Assist

- Structured SOAP note editor: Subjective (chief complaint, history), Objective (vitals, exam findings), Assessment (diagnosis), Plan (treatment).
- AI auto-populate: Doctor can speak into a microphone or paste a rough note. AI reformats it into a clean SOAP structure with proper medical terminology.
- ICD-10 code auto-suggestion: AI suggests ICD-10 codes based on the diagnosis text. Doctor selects with one click.
- Clinical decision support: AI flags if a prescribed drug contradicts a known allergy or current medication.

Orders

- Lab orders: Search test catalog, add multiple tests, mark urgency. AI suggests relevant tests based on diagnosis.
- Imaging orders: X-ray, MRI, CT scan with body region selection and clinical indication.

- Pharmacy orders: Drug name search with auto-complete from formulary, dosage, frequency, duration, SIG instructions.
- Referral orders: Refer to another specialist within the same hospital or external, with reason.
- IPD admission request: Pre-fill admission notes from consultation, select ward preference.

Vitals Entry

- Quick vitals form: BP, Pulse, SpO2, Temperature, Weight, Height, BMI (auto-calculated), RBS.
- Vitals trend chart embedded in consultation view.
- IoT integration ready: API endpoint to receive vitals from connected devices (pulse oximeters, BP machines with Bluetooth/WiFi).

6.3 Doctor Schedule Management

- Calendar view of all upcoming appointments. Block time for personal tasks.
- Set availability templates: recurring weekly availability, holiday blocking.
- Waiting room view: see who is physically checked in and ready.
- After-hours telemedicine slot management.

6.4 Doctor Analytics

- Patients seen: daily, weekly, monthly trend.
- Most frequent diagnoses (ICD-10 distribution).
- Avg consultation duration. Identify bottlenecks.
- Patient satisfaction score from post-visit feedback.
- Revenue generated from consultations.

7. OPD Management

7.1 AI Triage Kiosk — Upgraded

The existing triage system must be significantly enhanced:

- Multi-language support: Hindi, Tamil, Telugu, Bengali for triage questions.
- Voice input: patient can speak their symptoms. Speech-to-text (Whisper API) transcribes and AI processes.
- Symptom checker with body diagram: patient taps on a body diagram to indicate pain location.
- Triage output now generates: priority level, suggested department, estimated wait time, and preparation instructions (e.g., come fasting for blood sugar test).
- Auto-assign to available doctor in the suggested specialty. If no doctor available, queue for when one becomes available with patient notification.

7.2 Reception Dashboard

- Walk-in registration: Quick 30-second registration form. Only name, phone, DOB required. Rest filled later.
- Appointment check-in: Search by name or appointment ID. One-click check-in that moves patient to doctor queue.
- Token generation: Print or WhatsApp-send queue token to patient on arrival.
- Multi-queue management: Track queues across all active doctors simultaneously.
- Waiting room display: A separate screen mode showing current tokens being called across departments.

7.3 OPD Visit Lifecycle

1. Arrival	Patient arrives or confirms appointment. Check-in at reception or self-service kiosk.
2. Triage	AI or nurse collects symptoms and vitals. Assigns priority and department.
3. Queued	Patient token added to doctor's queue. WhatsApp notification: 'You are 3rd in queue, estimated wait: 20 min.'
4. Consultation	Doctor sees patient, writes SOAP note, creates orders.
5. Orders fulfilled	Lab goes to lab dashboard. Pharmacy goes to pharmacy. Billing auto-updated.
6. Results Review	Doctor reviews results when available. Updates treatment plan.
7. Billing	Final bill auto-generated. Patient pays at counter or online via portal.
8. Discharge	OPD discharge summary generated. WhatsApp prescription and follow-up instructions sent.

8. IPD Management

8.1 Bed Matrix — Enhanced

- Visual ward floor map: drag-and-drop bed assignment (optional premium feature).
- Real-time occupancy across all wards on one screen. Filter by ward type.
- Bed turnover tracker: time since last discharge, time to predicted next vacancy.
- Housekeeping integration: When a patient is discharged, bed status moves to 'Cleaning Required'. Housekeeping staff marks it clean, then it becomes Available.

8.2 Admission Workflow

- Admission from OPD: Doctor clicks 'Admit Patient' in their panel. Pre-populates admission form with diagnosis and notes.
- IPD Manager assigns bed based on ward preference, bed type (general/private/ICU), and gender.
- Admission document: auto-generate admission note PDF sent to patient's portal.
- Attendant/Guardian registration: Name, contact, relationship. Gets WhatsApp updates on patient status.
- Admitting package: if hospital offers packages (e.g., Normal Delivery Package ₹25,000), link to billing.

8.3 Daily Rounding

- Nursing notes: shift-wise nursing observations (vitals every 4 hours, medication administration record, IV fluid input/output chart).
- Doctor rounds: doctor visits patient, updates clinical progress notes, modifies orders.
- Multi-disciplinary team (MDT) notes for ICU patients.
- Critical alerts: if SpO2 drops below threshold or BP goes critical, auto-alert the attending doctor via WhatsApp and in-app.

8.4 Discharge Management

- Discharge planning: predicted discharge date set during admission for capacity planning.
- Discharge checklist: pending lab results, pending medications, outstanding bills — all must be cleared before discharge.
- Discharge summary auto-generation: AI compiles the full IPD stay into a structured discharge summary with diagnosis, treatment, medications, and follow-up plan.
- Post-discharge WhatsApp message: discharge instructions, medication schedule, follow-up appointment booking link.
- Readmission flag: if patient is readmitted within 30 days, system flags it and shows previous admission context immediately.

9. Laboratory Management

9.1 Lab Order Flow

14. Doctor creates lab order from consultation view. Order goes to lab dashboard instantly.
15. Lab technician sees order queue sorted by urgency. Prints sample collection labels.
16. Sample collected and marked. Order status changes to 'Sample Collected'.
17. Tech enters results. System compares against reference ranges and auto-flags abnormal values.
18. Results reviewed and validated by senior technician or pathologist.
19. Results auto-released to: doctor's dashboard, patient portal. WhatsApp sent to patient.

9.2 AI-Powered Lab Features

- Smart report narrative: AI generates a plain-language interpretation paragraph attached to every lab report. 'Your hemoglobin is slightly below normal range, which may suggest mild anemia. Please discuss with your doctor.'
- Critical value alerting: If a result is dangerously out of range (e.g., K+ > 6.5), immediate WhatsApp alert to doctor + a pop-up on doctor dashboard.
- Trending analysis: AI identifies when a biomarker shows a concerning trend across multiple visits even if each individual value is in range.
- Test bundle recommendations: Based on diagnosis, AI suggests commonly ordered test panels to improve efficiency.

9.3 Lab Inventory & Equipment

- Reagent inventory: Track reagent stock levels. Low stock alerts for procurement.
- Equipment maintenance log: record maintenance dates, breakdowns, calibration records for regulatory compliance.
- Outsourced tests: For tests the lab can't run, send to reference lab. Track outsourced orders with ETA.
- QC (Quality Control) logs: Daily QC run results to ensure machine accuracy.

10. Pharmacy Management

10.1 Smart Dispensing Workflow

20. Doctor creates prescription from consultation panel. Instantly appears on pharmacy dashboard.
21. Pharmacist views prescription with patient allergies and current medications highlighted.
22. Drug interaction check: automatic check against all current patient medications. Alert if interaction found.
23. Pharmacist confirms availability, substitutes if generic equivalent preferred.
24. Medication dispensed, status updated. Billing auto-updated.
25. WhatsApp message to patient: 'Your medication is ready for collection at pharmacy counter 2.'

10.2 Inventory Management

- Medicine master catalog: drug name (generic + brand), category, form (tablet/syrup/injection), HSN code for GST.
- Batch tracking: every batch has batch number, expiry date, MRP, purchase price. FIFO dispensing enforced.
- Reorder alerts: configure minimum stock level per drug. System auto-generates purchase order draft when stock falls below threshold.
- Expiry management: drugs expiring in < 30 days highlighted in amber. Expired drugs blocked from dispensing.
- Narcotic register: controlled substances have a separate digital register maintained for regulatory compliance.

10.3 Advanced Features

- Patient medication history: pharmacist can see all prescriptions ever dispensed to this patient across all visits.
- Formulary management: hospital can restrict prescribing to approved drug list to control costs and standardize care.
- Return and wastage tracking: record medication returns, breakages, or wastage with reason.
- Supplier management: preferred supplier per drug, purchase order management, GRN (Goods Received Note) processing.

11. Finance & Billing

11.1 Real-Time Billing Engine

All charges flow into the patient's invoice automatically as they are generated:

Consultation	Auto-added when doctor marks consultation complete. Uses doctor's configured fee.
Lab Tests	Auto-added when lab order is placed. Uses lab test price from master.
Pharmacy	Auto-added when prescription is dispensed. Uses batch-specific MRP.
IPD Room	Daily room charge calculated based on admitted bed type. Added at midnight each day.
Procedures	Surgeries, endoscopies, physiotherapy sessions added by treating department.
Miscellaneous	Any other charges added manually by finance with reason code.

11.2 Invoice & Payment

- Invoice displays a clean itemized bill grouped by category. Shows GST breakdown where applicable.
- Partial payment support: patient pays part now, rest on discharge.
- Multiple payment modes per invoice: split ₹5,000 UPI + ₹5,000 cash recorded separately.
- Advance payment: patient can pay advance on admission. Adjusted in final bill.
- Insurance billing: separate insurance claim flow. Pre-auth request, claim submission, TPA coordination, final settlement.
- Online payment via Razorpay on patient portal with real-time payment status via webhook.
- GST-compliant invoices with GSTIN, SAC codes. PDF invoice generation.

11.3 Financial Reporting

- Daily collection report: cash, UPI, card, insurance broken down by department.
- Outstanding dues: patients with unpaid balances older than 7, 30, 90 days.
- Department revenue: revenue attributed per department/doctor.
- Cost of goods sold: pharmacy margin analysis (purchase price vs billed price).
- MIS reports: Monthly management information system report exportable to Excel.
- AI financial insights: AI identifies unusual spikes, low-revenue departments, seasonal patterns.

12. Notification System — WhatsApp + Email

12.1 WhatsApp Business API Integration

Use Meta's official WhatsApp Business API (Cloud API) — not third-party providers. This ensures message delivery reliability and compliance.

- Store API credentials (phone number ID, access token) per organization in OrganizationConfig.
- All outgoing messages are template-based (approved by Meta). Templates stored in DB with variables.
- Message delivery status webhook: Delivered, Read, Failed statuses updated in Notification table.
- Build a WhatsApp template builder in the admin panel so hospital admins can create and submit new templates for Meta approval.

WhatsApp Notification Events

Campaign invite	AI health assessment invitation with unique link. Personalized with patient name.
Appointment confirmation	Booking confirmed. Date, time, doctor name, directions.
Appointment reminder	24h and 1h before appointment.
Lab result ready	Your lab results are ready. View them here: [link]
Prescription ready	Your medication is ready at pharmacy.
Bill generated	Bill summary. Pay online: [link]
Payment received	Payment confirmation with receipt number.
Discharge summary	You have been discharged. Download your summary: [link]
Critical lab alert	URGENT: Your [test] result requires immediate attention. Contact your doctor.
IPD progress update	Daily summary to patient's guardian: Dr. X reviewed [Patient] today. Condition: stable.

12.2 Email Notifications

- Nodemailer with per-org SMTP configuration (admin sets up their own SMTP or uses platform default SendGrid).
- HTML email templates using React Email library for beautiful, responsive emails.
- Emails used for: registration welcome, password reset, appointment details (with .ics calendar attachment), discharge summary (PDF attached), invoice (PDF attached).

13. AI Integration Strategy

13.1 AI Services Architecture

AI Service Layer Design
Create a dedicated app/lib/ai-service.ts module that abstracts all AI provider calls.
Support multiple providers: OpenAI (GPT-4o), Anthropic (Claude), and a local rule-engine fallback.
Per-org AI key: if org provides their own OpenAI key in config, use it. Else use platform key. This isolates costs.
Rate limiting: per-org AI request limits to prevent abuse. Track usage in AiUsageLog table.
Streaming support: for long AI responses (health summary, discharge notes), use streaming to show progressive output.
Prompt versioning: store all system prompts in DB with version history. Easy to update and A/B test without code deploy.
AI response caching: cache identical AI responses (same symptoms + same patient history hash) for 1 hour to save costs.

13.2 AI Use Cases — Full Catalog

AI Use Case	Description & Technical Approach
Health Assessment	Conversational questionnaire → structured summary, risk level, test recommendations. GPT-4o with function calling.
AI Triage	Symptoms + vitals → priority level + department assignment. Replaces rule-based triage.
SOAP Note Assistant	Doctor rough notes → formatted SOAP note with proper terminology. Voice input supported.
ICD-10 Coder	Diagnosis text → suggested ICD-10 codes. Fine-tuned on medical coding datasets.
Clinical Decision Support	Prescription → drug interaction check + allergy check. Uses OpenFDA + AI verification.
Lab Interpretation	Lab values + reference ranges + patient context → plain-language explanation.
Discharge Summary	Full IPD stay data → structured discharge summary. Doctor reviews and approves.
Financial Anomaly Detection	Daily revenue data → AI flags unusual patterns, potential billing errors.

Appointment Scheduling Optimizer	Queue data + doctor availability → optimal appointment slot suggestions.
Patient Risk Stratification	Patient history → chronic disease risk scores. Flags high-risk patients for proactive outreach.

14. Tech Stack Upgrades

14.1 Must-Add Dependencies

Package	Purpose
@prisma/client + prisma	Upgrade to latest. Add multi-tenant middleware.
next-auth v5 (beta)	Replace custom JWT with NextAuth. Supports multiple providers, sessions, and 2FA hooks.
openai	Official OpenAI SDK for all AI features.
@anthropic-ai/sdk	Claude SDK as AI fallback option.
react-email + nodemailer	Beautiful HTML email templates with server-side rendering.
zod	Schema validation for all Server Actions and API inputs. No unvalidated data enters DB.
uploadthing	File uploads (profile photos, medical documents, lab images) with S3-compatible storage.
@daily-co/daily-js	WebRTC for telemedicine video calls. Embeds in patient portal and doctor panel.
jspdf + html2canvas	PDF generation for invoices, discharge summaries, lab reports client-side.
date-fns	Date manipulation for appointment scheduling, shift management.
recharts	Charts for analytics dashboards, lab result trends, financial reporting.
zustand	Lightweight client-side state management for complex UI flows (consultation view, IPD matrix).
react-query (tanstack query)	Server state management. Handles caching, background refetch for live dashboards.
socket.io or Pusher	Real-time updates: queue changes, bed matrix updates, critical alerts.
bull + redis	Background job queue for: sending notifications, generating PDFs, running AI assessments.
sharp	Server-side image optimization for uploaded patient photos.

14.2 Database Migration: SQLite → PostgreSQL

- SQLite must be replaced with PostgreSQL for production. It cannot handle multi-tenant concurrency.
- Use Supabase (managed PostgreSQL with built-in auth, storage, and realtime) or Neon.tech (serverless PostgreSQL) for production.

- Enable Row Level Security (RLS) in PostgreSQL with org-scoped policies as a second line of defense.
- Connection pooling with PgBouncer (included in Supabase) is mandatory to handle hundreds of concurrent hospital connections.
- Migration script: write a data migration script to move existing SQLite data to PostgreSQL without data loss.

14.3 Infrastructure & Deployment

- Deploy on Vercel (Next.js optimized) or a dedicated VPS with PM2 + Nginx. Vercel recommended for ease.
- Environment management: use Vercel environment variables or Doppler for secrets management.
- CDN: Cloudflare for all static assets and as DDoS protection layer.
- File storage: Cloudflare R2 (S3-compatible, free egress) for all uploaded documents and images.
- Redis: Upstash Redis (serverless) for Bull job queue and API rate limiting.
- Monitoring: Sentry for error tracking, Posthog for product analytics, Uptime Robot for uptime monitoring.
- CI/CD: GitHub Actions for automated testing and deployment pipeline.

15. Security & Compliance Architecture

15.1 Authentication & Authorization

- Implement RBAC (Role-Based Access Control) with fine-grained permissions. Each role has a permission set stored in DB.
- Permission check utility: checkPermission(user, action, resource) called at the start of every Server Action.
- All auth tokens are short-lived (15 min access, 7 day refresh). Refresh tokens stored in httpOnly cookies.
- 2FA enforcement: mandatory for admin and doctor roles. TOTP (Google Authenticator / Authy).
- Session management: view and revoke active sessions from profile settings.
- Login anomaly detection: alert if login from new device or unusual location.

15.2 Data Security

- Encryption at rest: all sensitive fields (Aadhaar, insurance numbers, medical history) encrypted in DB using AES-256.
- Encryption in transit: TLS 1.3 enforced. HSTS headers.
- PII masking in logs: Sentry and application logs must never log patient names, phone numbers, or medical data.
- HIPAA-aligned data handling: business associate agreements with all third-party services (OpenAI, AWS, etc.).
- Data retention policies: configurable per org. Patient data archived after X years of inactivity.
- Right to deletion: patient can request data deletion from portal. Triggers a soft-delete with a 30-day grace period.

15.3 Audit & Compliance

- Every clinical action (create/update/delete on EHR, prescriptions, orders) writes to system_audit_logs.
- Audit log includes: who, what, when, from which IP, old value, new value.
- Audit logs are append-only — no staff member at any level can delete or modify audit logs.
- Compliance reports: admin can export audit logs for any date range as CSV for regulatory inspection.

16. AI Agent Implementation Roadmap

This section is the direct instruction set for your AI agent. Execute phases sequentially. Each phase builds on the previous. Do not skip phases.

Phase 1 — Foundation & Multi-Tenancy

Timeline: Weeks 1-3

- Upgrade Prisma schema: Add Organization, OrganizationConfig, OrganizationBranding models.
- Add organizationId to all existing models. Write and run migration.
- Build Prisma middleware for automatic tenant scoping on all queries.
- Replace custom JWT with NextAuth v5. Configure credential + OTP providers.
- Build Super Admin panel: /superadmin route with its own auth, hospital list, add hospital wizard.
- Update middleware.ts: Add superadmin route protection and org subdomain routing.
- Write comprehensive Zod schemas for all existing Server Actions. Add input validation.
- PostgreSQL migration: Set up Neon or Supabase, migrate schema, configure connection pooling.

Phase 2 — Patient Portal & AI Assessment

Timeline: Weeks 4-6

- Build patient portal frontend: /patient route rewrite. Mobile-first, glassmorphism UI.
- Implement phone OTP authentication for patients using Twilio or MSG91.
- Build AiHealthAssessment table and assessment session management.
- Build the conversational assessment UI: multi-step with dynamic questions, body diagram, voice input.
- Build AI service layer (app/lib/ai-service.ts) with OpenAI integration and rule-engine fallback.
- Connect assessment answers → GPT-4o → structured health summary output.
- Build patient dashboard: upcoming appointments, lab results, prescriptions, payments.
- Integrate UploadThing for patient document uploads.
- Build appointment slot system: AppointmentSlot model, slot generation, slot booking UI.

Phase 3 — Notification System & Communications

Timeline: Weeks 7-8

- Integrate WhatsApp Business Cloud API. Build message sending utility in app/lib/whatsapp.ts.

- Build notification template system in DB. Admin panel UI to manage templates.
- Implement all 10 WhatsApp notification events listed in Section 12.
- Build React Email templates for all email notification types.
- Set up Bull + Upstash Redis for notification job queue (never send notifications synchronously).
- Build campaign manager in hospital admin panel: segment patients, send assessment invites.
- Implement delivery status webhooks and update Notification table.
- Build notification preferences: patient can opt-out of specific notification types.

Phase 4 — Doctor Panel AI Augmentation

Timeline: Weeks 9-11

- Rebuild doctor queue with real-time updates via Pusher or Socket.io.
- Build AI pre-consultation brief: auto-generate patient summary when consultation is opened.
- Implement AI SOAP note assistant: voice input with Whisper API + AI reformatting.
- Add ICD-10 auto-coder to diagnosis field using AI.
- Build clinical decision support: drug interaction checker on prescription creation.
- Add AI lab test bundle suggestions to order creation flow.
- Build telemedicine video call UI using Daily.co WebRTC SDK.
- Implement doctor analytics dashboard with recharts.

Phase 5 — IPD, Lab, Pharmacy, Finance Polish

Timeline: Weeks 12-14

- IPD: Build interactive bed matrix with color coding and real-time updates.
- IPD: Implement daily nursing notes, rounding notes, MDT notes.
- IPD: Build AI discharge summary generator. Doctor review + approve flow.
- Lab: Add AI lab report interpretation paragraph to every result.
- Lab: Implement critical value alerting via WhatsApp to doctor.
- Pharmacy: Build drug interaction checker at point of dispensing.
- Finance: Build real-time billing engine with auto-charge triggers.
- Finance: Implement Razorpay online payment on patient portal with webhook.
- Finance: Build all financial reports with Excel export.

Phase 6 — Security, Compliance & Production Hardening

Timeline: Weeks 15-16

- Implement full RBAC with fine-grained permission system.
- Add AES-256 encryption for all sensitive fields in DB.
- Implement 2FA (TOTP) for admin and doctor roles.

- Review and complete audit logging on every clinical and financial mutation.
- Add PII masking to all logging and error tracking configurations.
- Security audit: run OWASP ZAP scanner on all routes. Fix all critical and high findings.
- Load testing: use k6 or Artillery to simulate 500 concurrent users. Identify and fix bottlenecks.
- Set up Sentry error monitoring, Posthog analytics, Uptime Robot alerting.
- Write end-to-end tests for critical flows: patient registration → consultation → billing → payment.
- Complete API documentation for all Server Actions. Prepare deployment runbook.

17. High-Impact Quick Wins (Do These First)

If you need to prioritize for an MVP demo or investor presentation, these features deliver maximum visible value with minimum effort:

#	Feature	Why It Matters
1	Multi-hospital super admin panel with add hospital wizard	Demonstrates the SaaS business model. Investors immediately see the scalability story.
2	AI Health Assessment with WhatsApp invite link	Most novel and differentiated feature. No other HMS does this. Perfect demo moment.
3	Patient portal with appointment booking and online payment	Patients love self-service. Hospitals love the reduced reception load.
4	Real-time OPD queue with token notifications to patient's WhatsApp	Visually impressive and practically useful. Reduces crowding in waiting rooms.
5	AI SOAP note assistant with voice input	Doctors love anything that reduces documentation time. High satisfaction score.

Final Note to AI Agent
This document is your complete north star for the Hospital OS Enterprise upgrade.
Always enforce multi-tenancy at every layer — a single data leak across tenants is catastrophic.
Prioritize patient data security above every other concern.
Ship Phase 1 (multi-tenancy foundation) before writing a single line of new feature code.
Every Server Action must be validated with Zod and authorized with a permission check — no exceptions.
Write tests for every critical path: auth, billing, prescription dispensing, and lab results.
This plan is comprehensive — you now have everything you need. Build something great.

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